

JOSHWYN PAREKH

Computer Engineer @ University of Waterloo

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering, Honours, Co-op

Waterloo, ON

Sep. 2022 – Present

SKILLS

Languages: C, C++, C#, Python, HTML5/CSS, JS, Verilog, MATLAB, VHDL, ARMv7, Kotlin, $\LaTeX$

Developer Tools & Frameworks: Unity, UE5, Blender, Power Automate/Apps, AWS, MongoDB, Docker, Git

Design & Editing: AutoCAD, 3ds Max, Revit, Tinkercad, Shotcut, Canva, Camtasia, Synthesia

Interpersonal Skills: Lvl-1 tennis and ski coach, strengthened critical thinking, collaboration, and communication skills

EXPERIENCE

Virtual Reality Game Developer

Liftwerx

May. 2025 – Aug. 2025

Cambridge, ON

- Designed and developed interactive VR training experiences in Unity using C#, applying game development principles to create immersive, gamified work procedures for the LiftWerx VR Academy training program.
- Built and refined interactable scripts, 3D game assets and environments using Blender, SolidWorks, and Unity, supporting fast-paced prototyping and production of polished, gameplay-driven VR content for on-site technicians.

Systems Software Developer

Co-operators

Sep. 2024 – Dec. 2024

Guelph, ON

- Led the end-to-end development of a company-wide automated notification system, independently managing all stages of the software development life cycle from design and implementation to deployment and maintenance.
- Integrated multiple enterprise applications using APIs, secure authentication and web services to enable seamless communication for data collection within the automated systems.
- Built an AWS-based chatbot using the ChatGPT API, HTTP requests, and Lambda functions to query a database to be able to dynamically retrieve and validate data to deliver accurate responses.

PROJECTS

Custom Operating System Kernel | C, ARM v7

Jan. 2025 – Apr. 2025

- Developed a lightweight operating system kernel with a preemptive, priority-based scheduler, dynamic memory allocation, and synchronization primitives.
- Established process creation (running other C files), context switching, and inter-process communication, integrating C with Assembly for low-level hardware/memory interaction.

LLM Hardware Block Implemented on an FPGA | SystemVerilog, Pynq Board

Jan. 2025 – Apr. 2025

- Implemented the BERT transformer model (a core architecture behind modern Large Language Models) using integer-only operations which learns word relationships and context.
- Built all key modules (quantization, softmax, normalization, GELU) in SystemVerilog at the RTL level, and deployed the model on a Pynq FPGA board achieving highly efficient hardware execution.

Full Stack Statistics Tracker for *The Finals* | Python, Tkinter, Pytesseract, Pyautogui

Oct. 2024 – Dec. 2024

- Developed a full stack Python application to display user statistics from different gaming sessions in 'The Finals'.
- A screenshot of the in-game scoreboard is captured at the end of each game and individual statistics can be viewed via text extraction using an OCR. All data is also parsed and then stored in a database for future retrieval.

“ TESTIMONIALS

The following testimonials highlight my performance across various co-op positions I've completed during my degree at the University of Waterloo. Each testimonial is from my direct manager, all of whom rated my work terms according to the Co-op student performance evaluation guide, as **OUTSTANDING**.

André Larocque

LiftWerx

Rating: Outstanding

“Joshwyn exceeded expectations in terms of the quality of his work, the quantity of valuable output, and his contributions to projects beyond the scope of his role. . . You are a capable, talented, intelligent developer and designer. Your critical insights and the reliability of your word are invaluable to any workplace.”

- Recognized as a talented, reliable developer and collaborator.
- Exceeded expectations in quality, quantity, and impact of contributions.
- Provided critical insights that shaped project planning and team success.

Linda Nguyen

The Co-operators

Rating: Outstanding

“Joshwyn was great with driving all of his tasks to completion and self-managed the various tasks. Documentation was concise and well structured. His presentation and communication skills were praised. . . Overall, Joshwyn was great to work with and it makes me very open to considering hiring another student in the future if they're anywhere near Joshwyn's dedication.”

- Independently drove tasks to completion with minimal oversight.
- Delivered high-quality automations, documentation, and presentations.
- Excellent adaptability, feedback incorporation, and collaboration.

Maureen Barrow

Dematic Logistics Systems

Rating: Outstanding

“Joshwyn is a highly valued member of our team, and his contributions have not gone unnoticed. His exceptional work ethic and positive attitude have made him a pleasure to work with. . . Overall, Joshwyn's contributions to our team have been invaluable, and we would be fortunate to have him return as a full-time member. Joshwyn is a terrific employee. I would rehire him in a heartbeat.”

- Highly valued team member with exceptional work ethic and professionalism.
- Strong presentation and communication skills; repeatedly commended by colleagues.

All testimonials provided by direct supervisors during end of term co-op performance evaluations.